

Technical Report Gap Analysis & Fix Report

Date: 2026-03-16

Scope: Comprehensive audit of `docs/technical_report.md` — figures, captions, and image references

Status: All issues resolved

Issues Found

Issue 1: Duplicate Images in Figure 4 Section (§5.7 CBD Robustness)

Problem: Two images were displayed in the Figure 4 area:

1. **Unlabeled** `![CBD Scenario Comparison](/home/ubuntu/ems-optimization/results/figures/cbd_scenario_comparison.png)` — a summary bar chart from EDA
2. **Labeled** `![CBD Robustness Enhanced](/home/ubuntu/ems-optimization/results/figures/cbd_robustness_enhanced.png)` — the actual Figure 4

The unlabeled image appeared immediately before the Figure 4 introduction paragraph, creating confusion about which image was Figure 4.

Fix: Removed the unlabeled `cbd_scenario_comparison.png` reference. Figure 4 now shows only `cbd_robustness_enhanced.png` with its caption.

Issue 2: Duplicate Images in Figure 5 Section (§5.11 CBD-Focused vs Manhattan-Wide)

Problem: Two images were displayed in the Figure 5 area:

1. **Unlabeled** `![CBD-Focused Comparison](/home/ubuntu/ems-optimization/results/cbd_focused_comparison/cbd_focused_comparison.png)` — a raw comparison chart from the analysis subdirectory
2. **Labeled** `![CBD Equity-Efficiency Tradeoff Summary](/home/ubuntu/ems-optimization/results/figures/cbd_equity_tradeoff_summary.png)` — the actual Figure 5

Fix: Removed the unlabeled `cbd_focused_comparison.png` reference. Figure 5 now shows only `cbd_equity_tradeoff_summary.png` with its caption.

Issue 3: Duplicate Images in Figure 6 Section (§5.12 Capacity Constraints)

Problem: Two images were displayed in the Figure 6 area:

1. **Unlabeled** `![Full Spectrum Capacity Summary](/home/ubuntu/ems-optimization/results/capacity_comparison/full_spectrum_summary.png)` — a multi-panel summary from the capacity comparison subdirectory
2. **Labeled** `![Capacity Sensitivity Heatmap](/home/ubuntu/ems-optimization/results/figures/capacity_sensitivity_heatmap.png)` — the actual Figure 6

Fix: Removed the unlabeled `full_spectrum_summary.png` reference. Figure 6 now shows only `capacity_sensitivity_heatmap.png` with its caption.

Issue 4: Figure 6 Caption Text Error — Incorrect K Value

Problem: The Figure 6 caption stated:

“At K=20 **and K=30**, capacity is non-binding for all policies...”

The heatmap actually shows **K=20 and K=40** (not K=30). K=30 data exists in supplementary tables but is not displayed in this figure.

Fix: Updated caption to:

“At K=20, capacity is non-binding for all policies...”

Removed the incorrect K=30 reference since the heatmap panels are K=20 and K=40 only.

Issue 5: Capacity Sensitivity Heatmap — “Data Format Issue”

Problem: The `capacity_sensitivity_heatmap.png` displayed blank panels with “Data format issue” text instead of actual heatmap data. This was caused by a policy name mapping issue — the CSV data used labels like `P0`, `P1_demand`, and `P2_optimised`, but the plotting script’s `POLICY_LABELS` dictionary wasn’t mapping all variants correctly in an earlier version.

Fix: Regenerated the heatmap by running `scripts/generate_capacity_sensitivity_heatmap.py`. The script now correctly maps all policy label variants and produces a proper heatmap with numerical values for all 3 policies × 5 capacity levels × 2 K values.

Verification Summary

Final Figure Inventory (inline figures)

Figure	Image File	Caption Present	Unique
Figure 1	results/figures/policy_comparison_panel_K20_cap2.png	Yes	Yes
Figure 2	results/figures/response_time_distribution_by_policy.png	Yes	Yes
Figure 3	results/figures/fleet_sensitivity_dual.png	Yes	Yes
Figure 4	results/figures/cbd_robustness_enhanced.png	Yes	Yes
Figure 5	results/figures/cbd_equity_tradeoff_summary.png	Yes	Yes
Figure 6	results/figures/capacity_sensitivity_heatmap.png	Yes	Yes

Supporting (unnumbered) Images

Section	Image File	Purpose
§5.8 Queueing	results/figures/queue_comparison_by_policy.png	Supplementary
§5.9 Seasonal	results/figures/seasonal_patterns.png	Supplementary
§5.10 Distance	results/distance_comparison/distance_comparison_bar.png	Supplementary

Image File Existence Check

All 9 referenced images verified to exist on disk with non-zero file sizes.

Changes Made

1. **Removed** 3 duplicate/unlabeled image references (lines ~528, ~609, ~660 in original)
2. **Fixed** Figure 6 caption: removed incorrect “K=30” reference
3. **Regenerated** `capacity_sensitivity_heatmap.png` with correct data rendering
4. Net line change: -6 lines (1161 → 1155)